UU-P-228c
August 12, 1965
SUPERSEDING
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April 11, 1958

FEDERAL SPECIFICATION

PAPER SET, DUPLICATING MASTER SPIRIT OR LIQUID PROCESS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers duplicating master paper sets for use in the spirit or liquid process.

1.2 Classification.

1.2.1 *Types, grades, and classes.* The paper sets shall be of but one type and shall have the following grades and classes, as specified (see 6.2):

Class 1—Four-part set.
Class 2—Three-part set, fold over style.
Class 3—Three-part set, stub construction.
Grade A—Smudge resistant (carbon overcoated with protective film).
Grade B—Regular (carbon not overcoated with a protective film).

1.2.2 *Sizes.* The paper sets shall be of the size specified in the invitation for bids (see 6.2). The size shall indicate the reproduction area of the master sheet.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification:

Federal Specifications:

O-D-825—Duplicating Fluid, Direct Process, (Spirit Type).
UU-P-31—Paper; General Specifications and Methods of Testing.
UU-P-150—Paper, Carbon, Duplicating Machine, Spirit-Process.
UU-P-232—Paper, Duplicating Copy, Liquid Process.
UU-P-233—Paper, Duplicating, Master, Spirit-Process.
UU-P-258—Paper, Index.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Master sheets.* Master sheets shall conform to the requirements of UU-P-233.

3.1.2 *Duplicating (carbon) paper.*

3.1.2.1 *Grade A sets.* For grade A, class 1, 2, or 3 paper sets, the carbon paper shall conform to the requirements of type I, grade A of UU-P-150.

3.1.2.2 *Grade B sets.* For grade B, class 1, 2, or 3 paper sets, the carbon paper shall conform to the requirements of type I, grade B of UU-P-150.

3.1.3 *Protective sheet.* The protective sheet shall be nonabsorbent to the extent that the time required for penetration of a drop of castor oil shall be not less than 45 seconds when tested as specified in 4.2.1.

3.1.4 *Backing sheet (class 1).* The backing sheet may be of the same color, stock, and weight as the master sheet. If not the same paper as the master, the backing sheet shall be made of stock, free from unbleached or groundwood pulp having an average basis weight of not less than 51 pounds (25 x 38-500 sheets) when tested as specified in 4.2.1. The color shall be basically white.

3.2 Construction.

3.2.1 *Class 1, four-part set.* Class 1, grades A and B sets shall consist of a master sheet, a protective sheet, a duplicating sheet of the applicable grade (see 3.1.2), and a backing sheet. The duplicating sheet (carbon paper) shall be adhered to the backing sheet along the top edge and shall have the coated side facing the master sheet. The master sheet shall be perforated along the top edge, when the backing sheet is made of the same paper as the master, or along the top at a point 1/2 ± 1/8 inch from the top edge when a stub construction is used to join the master and backing sheets. The backing sheet shall extend to the bottom of the master sheet. The carbon paper shall be the width of the master sheet and shall extend to within 1/8 ± 1/16 inch of the bottom of the master sheet or alternatively may have the bottom 1/4 ± 1/8 inch, without carbon coating. The protective sheet shall be inserted loosely between the carbon sheet and the master sheet, and shall cover the entire area of the master sheet when aligned with the top edge, and extend 1/2 ± 1/8 inch beyond the bottom edge of the master sheet.

3.2.2 *Class 2, three-part set, fold over style.* Class 2, grades A and B sets shall consist of a master sheet, protective sheet, and a duplicating carbon sheet of applicable grade (see 3.1.2). The master sheet shall be perforated across the top edge and shall fold over the top of the carbon sheet for a distance of 1/2 ± 1/8 inch and glued to the back thereof. The carbon paper shall be positioned with the carbon coating facing the master sheet, and shall have an uncoated margin 3/8 ± 1/8 inch along the bottom edge which shall extend beyond the bottom of the master. The protective sheet shall be loosely inserted between the carbon sheet and the master sheet and shall cover the entire area of the master and extend 5/8 ± 1/8 inch further.

3.2.3 *Class 3, three-part set, stub con-*

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struction. Class 3, grades A and B shall have the same construction as class 2 sets except that the master sheet shall be perforated across the width of the sheet 1/2 ± 1/8 inch from the top edge, and the carbon sheet shall be glued to the stub formed by the perforations and top edge of the master sheet.

3.3 Color. The color of the reproductions produced by master sets shall be black, blue, or purple, as specified (see 6.2).

3.4 Performance.

3.4.1 *Duplicating quality.* Type II duplicating master sets, prepared and tested as described in 4.3.2, shall be capable of reproducing the following number of copies for the color specified, in one continuous run, when the master is used immediately after it is prepared. All copies shall be legible, clean, have sharply defined letters, and shall be of good color.

Color	Number of copies
Black	100
Blue	300
Purple	300

3.4.2 *After one-hour drying.* Type II, duplicating master sets, prepared as specified in 4.3.2, shall reproduce 50 clean, legible copies, and after allowing the master to dry for one hour, shall produce an additional 50 copies of equal quality when tested as specified in 4.3.2.

3.4.3 *Interrupted operation.* Type II, duplicating master sets prepared as specified in 4.3.2, shall reproduce 90 clean and legible copies after interrupted operation as specified in 4.3.2.

3.5 Dimensions. The paper sets shall be the width specified (see 6.2), with a tolerance of ± 1/16 inch. The master sheet shall be the length specified (see 6.2), plus a margin of no more than 1/2 inch for clamping the sheet in the reproduction equipment. The length available for typing or printing, on the master sheet, shall be as specified (see 6.2), with a tolerance of ± 1/8 inch.

3.6 Printing. Printing of duplicating master sets, when desired, shall be as specified in the contract or order (see 6.2).

3.7 Workmanship. Master sets shall be clean, uniformly coated, and free from pinholes, uncoated areas, or other defects which might affect the quality of reproduction.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities, or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1 *Sampling and testing of components.* The paper set backing sheet, when not the same paper as the master sheet, and the protective sheet shall be tested for the applicable characteristics in table I. The methods of testing specified in UU-P-31, where applicable, and as listed in table I, shall be followed. The lot shall con-

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TABLE I. *Instructions for testing components*

Characteristic	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as numerically to nearest[1]
	Requirement	Test method	Indiv. unit	Lot aver.		
Atmospheric conditions	4.2.1	102	All testing		—	—
Protective sheet						
Absorbency	3.1.3	150	X	—	5	nearest 5 seconds
Backing sheet						
Stock	3.1.4	800	—	—	2 on composite	5 percent
Basis weight	3.1.4	110	—	X	1 on 1000 square inch specimen	0.1 pound

[1] Test reports shall include all values on which results were based.

sist of all the backing paper and protective sheet paper as applicable, manufactured at one time, for use in the assembly of paper sets, and shall be expressed in pounds of paper. The sample unit shall consist of at least 1000 square inches of paper. The sample size or sample unit shall be selected in accordance with the sampling plan below, and there shall be no failure to meet the composite or lot average requirements. Test sample units shall be conditioned in accordance with method 102 of UU-P-31, and testing shall be made under the atmospheric conditions specified.

Lot size	Sample size
(*Pounds of paper*)	
800 or less	2
801 up to and including 22,000 ...	3
22,001 or more	5

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) in 4.3.1.4. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of paper sets of one grade and class for examination under 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination under 4.3.1.3.

4.3.1.1 *Examination of the end item for defects in appearance, construction, and workmanship.* The sample unit for this examination shall be one paper set. No more than one sample unit shall be selected from a package.

Examine	Defect
Form	Not grade or class specified.
Appearance	
Color	Duplicating sheet (carbon paper), not color specified.
	Backing sheet not basically white.
Construction	Protective sheet missing; not loosely inserted, adhered to adjacent sheets, does not cover entire area of master sheet.
	Coated side of carbon paper not facing master sheet; not same same width of master sheet.
	Master sheet not perforated to permit removal.
	Carbon paper not securely adhered to backing sheet, stub, or fold over portion of master sheet, as applicable.
Workmanship	Sets dirty, smudged; contains holes.
	Any sheet creased, torn, or wrinkled.

4.3.1.2 *Examination for dimensional defects.* The sample unit for this examination shall be one paper set.

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Examine	Defect
Width of set (all parts)	Not within tolerance for specified width.
Master sheet	Length of typing or printing area not within tolerance of specified size. Perforations less than 3/8 inch or more than 5/8 inch from top edge, when stub construction is used.
Protective sheet	Not longer than master sheet by 1/2 ± 1/8 inch.
Duplicating sheet (carbon paper)	Length of coated portion not within 1/8 inch, ± 1/16 inch of bottom of master sheet.

4.3.1.3 *Examination of preparation for delivery.* An examination shall be made to determine that markings, packaging, and packing comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. An equal number of shipping containers fully prepared for delivery shall be examined for closure defects. The lot shall be unacceptable if the average count per package is less than specified or indicated, when one package is examined for count from each open shipping container selected for the preparation for delivery examination.

Examine	Defect
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components; such as incomplete closure of container flaps, loose, or inadequate sealing, strapping, or stapling. Bulged or distorted containers.
Contents	Instruction sheet missing; not pressboard; not basis weight specified.
Weight	Net weight of contents exceeds requirements.

4.3.1.4 *Inspection levels and AQLs for examinations.* The inspection levels for determining the sample size and the AQLs expressed in defects per 100 paper sets or shipping containers, as applicable, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1.1	S-4	2.5
4.3.1.2	S-2	4.0
4.3.1.3	S-2	2.5

4.3.2 *Testing of the end item.* Paper sets shall be tested for the applicable characteristics in table II. The sample unit shall be three complete sets. The lot size for the purpose of determining the sample size shall be expressed in units of paper sets of one grade, color, and class. The sample size shall be selected in accordance with the sampling plan below. There shall be no failure to meet the applicable requirements specified in 3.4.

Lot size (*Pounds of paper*)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.4 **Tests.**

4.4.1 *Performance quality.*

4.4.1.1 *Apparatus.* The preparation of duplicating master sets shall be made on an electric typewriter with elite or pica type. The typewriter shall be equipped with a typewriter ribbon not more than 0.0050 inch thick and a platen having a Shore durometer reading of between 89 to 93 or an equivalent platen designed for the preparation of duplicating masters on the machine being used. The type shall be thoroughly cleaned before each test.

4.4.1.2 *Preparation of the master.* Duplicating master sets shall be placed in the typewriter as follows: For class 1 sets, the protective sheet is removed and the set inserted in the typewriter so that the backing

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TABLE II. *Instructions for testing end item*

Characteristic	Specification reference		Requirements applicable to individual unit	Number determinations per sample unit	Results reported as pass or fail
	Requirement	Test method			
Performance:					
Duplicating quality	3.4.1	4.4.1.3	X	1	X
After one hour drying	3.4.2	4.4.1.4	X	1	X
Interrupted operation	3.4.3	4.4.1.5	X	1	X

sheet is at the bottom nearest the platen and the master sheet on top. For classes 2 and 3 sets, the protective sheet is removed and the set inserted so that the duplicating (carbon) paper is on the bottom nearest the platen and master sheet is on the upper side to receive the type. The master sets shall then be typed with the entire keyboard, small letters, capitals, numerals, and characters twice in random order with the typewriter pressure set at one-fourth full pressure. The same typing shall then be repeated with the typewriter pressure set at one-half full pressure.

4.4.1.3 *Duplicating quality.* The typed master shall be detached from the set and placed on the duplicating machine. Duplicating fluid conforming to type II of O-D-825 shall be used in the duplicating machine, and the specified number of copies (see 3.4) shall be made on copy paper conforming to UU-P-232. For copies made from sets with a black carbon paper, the 25th, 50th, and 100th reproduced copies shall be examined for the duplicating quality specified in 3.4. For copies made from blue or purple duplicating paper, the 25th, 50th, 100th, 200th, and 300th copies shall be examined for conformance with the requirements of paragraph 3.4.

4.4.1.4 *After one-hour drying.* Prepare and set up a master sheet as specified in 4.4.1.2 and 4.4.1.3. Make 50 copies on the duplicating machine. Remove the master and allow to dry for one hour, then replace on the duplicator and run an additional 50 copies. Compare the quality of reproduction of 5 randomly selected copies of the initial run with 5 randomly selected copies of the run made after the dried master sheet has been reused.

4.4.1.5 *Interrupted operation.* Prepare and set up a master sheet as specified in 4.4.1.2 and 4.4.1.3. Make 10 copies on the duplicator, then continue running for 5 revolutions without feeding copy paper. Resume feeding copy paper until an additional 90 copies have been run off. Compare 10 randomly selected copies of the letter operation with the initial 10 copies for quality of reproduction.

4. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* One hundred duplicating master sets of one grade, class, and size only, shall be packaged in a snug-fitting setup paperboard box conforming to type I, grade A, class A, style 4 of PPP-B-676. Closure of the box shall be in accordance with the appendix of the box specification. Each package shall contain a reinforcing sheet of index paper, basis weight of 280 pounds (25-1/2 by 30-1/2—1000) conforming to grade C of UU-P-258; with complete instructions on the use of the duplicating paper printed thereon.

5.1.2 *Level C.* The duplicating master sets shall be packaged in accordance with the industry's practice. Each package shall

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contain a reinforcing sheet of index paper, basis weight of 280 pounds (25-1/2 by 30-1/2—1000) conforming to grade C of UU-P-258, with complete instructions on the use of the duplicating paper printed thereon.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Duplicating master sets of one grade, class, and size only, packaged as specified in 5.1, shall be packed in snug-fitting fiberboard shipping containers conforming to style RSC, V3c or V3s, of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 *Level B.* Duplicating master sets of one grade, class, and size only, packaged as specified in 5.1, shall be packed in snug-fitting fiberboard shipping containers conforming to style RSC, type CF or SF, class domestic of PPP-B-636. Closure shall be in accordance with the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.3 *Level C.* Duplicating master sets, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Duplicating master sets are intended for use in the spirit or liquid duplicating process. Grade A master sets are smudge resistant and minimize soiling of hands and clothing.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class, grade, and reproduction color required (see 1.2.1 and 3.3).
(*c*) Size of reproduction area required (see 1.2.2).
(*d*) Printing, when required (see 3.6).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 **Classification changes.** Type I sets have been deleted from this specification because the gelatin process is no longer used. Coverage is limited to the former type II sets.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Paper, not otherwise indexed by name, not printed.
Carload minimum weight, 36,000 pounds.

Motor:

Paper, not otherwise indexed, not printed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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